BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

The objective is to identify input parameters that maximize an unknown objective function using a limited number of evaluations. Success is measured by the best observed objective value and the efficiency of the search.
I adopted an iterative Bayesian optimisation framework, in which candidate input configurations were systematically evaluated and the resulting observations used to update a surrogate model of the underlying objective function. An acquisition function guided the selection of subsequent candidates, with a strategic emphasis on exploration over exploitation to improve coverage of the parameter space, reduce uncertainty, and identify potentially promising regions. Each successive observation contributed to refining the surrogate model, enabling progressively more informed decisions while mitigating the risk of premature convergence to local optima.

2. **How your strategy has evolved**
   - Describe the key changes in your approach since your early rounds.
   - What influenced these changes – data trends, feedback, model performance or intuition?
   - Identify which principles or heuristics now guide your query decisions.

My optimisation strategy primarily emphasised exploration over exploitation, with the objective of developing a comprehensive understanding of the underlying objective function and its response to different input configurations.
Rather than concentrating evaluations around regions exhibiting strong initial performance, I prioritised sampling across a broader parameter space to identify potentially unexplored regions with superior outcomes.
This approach was motivated by the need to reduce uncertainty, mitigate the risk of premature convergence to local optima, and improve the identification of promising regions within the search space.
Although this strategy may sacrifice short-term optimisation gains, it aims to improve the quality of information acquired during successive iterations and support more informed optimisation decisions.
My decisions increasingly rely on observed improvements, model uncertainty and the expected value of new evaluations. I favour candidates with either strong predicted performance or useful information potential

3. **Patterns, data and insights**
   - Highlight the most meaningful trends you've observed across your data points or model evaluations.
   - Identify which variables or behaviours seem to influence results the most.
   - Explain how these observations have shaped your understanding of the search process.

Across iterations, I would examine whether the best objective value improves steadily, plateaus or varies significantly. I would also compare observed outcomes with surrogate predictions to assess model reliability.
I would investigate which input variables appear most influential, including possible interactions and nonlinear effects. These patterns help narrow the search and identify where further sampling is valuable.

4. **Decision-making and iteration**
   - Explain how you balance exploration and exploitation.
   - Describe one or two examples of strategic decisions. What worked, what didn't, and why?
   - Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Exploration tests uncertain or under-sampled regions; exploitation focuses on inputs predicted to perform well. I balance the two using the acquisition function, adjusting the emphasis as evidence accumulates.
A useful decision is to sample a promising region after repeated improvement. A less effective choice may be to over-focus there and miss alternatives. When results are surprising, I update the model and reassess uncertainty.

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Next, I would test alternative acquisition settings, review parameter bounds and evaluate convergence against a simple baseline. I would also check whether the best solution remains robust under repeated or nearby evaluations.
This project illustrates how machine learning can guide decisions when evaluations are costly and the objective is unknown. For stakeholders, I would report the best result, improvement over baseline and remaining uncertainty in plain language.